Measuring Magnetic Field via Hall Effect

Magnetic field measurement is an important method to study the physical processes related to magnetic phenomena. In this experiment, Hall effect is used to measure magnetic field. In 1879, Hall, a graduate student at Hopkins University, found that if there is a current in a conductor and a magnetic field perpendicular to the direction of the current, an electric field perpendicular to both the direction of the current and the magnetic field will be generated in the conductor. This phenomenon is called Hall effect. The Hall effect has been used to make magnetic sensors to measure magnetic fields, but the Hall effect of metal is too weak to be used in practice. With the development of semiconductor materials and manufacturing technology, people use semiconductor materials to make Hall elements. Because of its remarkable Hall effect, it has been applied and developed. Now, Hall effect has been widely used in measurement technology, automation technology, computer and information technology.

Objectives

1. Understand the principle of Hall effect and parameters related to Hall element.
2. Learn to eliminate additional potential by symmetric measurement.
3. Calculate the sensitivity and carrier concentration of Hall element.
4. Learn to measure magnetic induction intensity and magnetic field distribution by Hall effect.

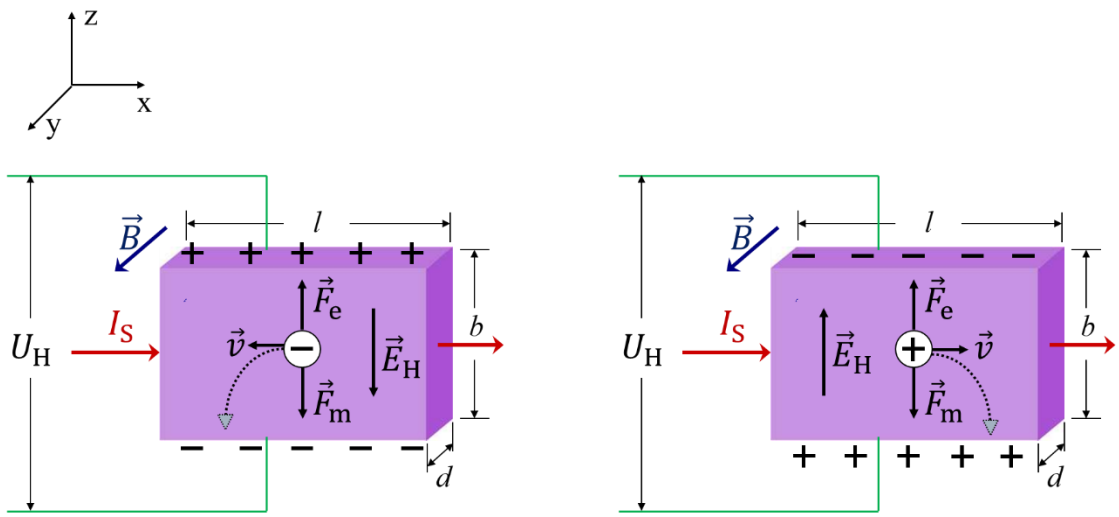
Apparatus

Hall effect tester, wires.

Theory

1.The Hall effect

Because the magnetic field can't be seen or touched, we can only use its properties and some of its effects to measure, Hall effect is one of them. To describe this effect, let’s consider a flat semiconductor (Hall element) with length, width and thickness of l, b and d respectively, as shown in Fig. 1. The current I_s (called working current) is in the direction of the +x-axis and there is a uniform magnetic field $\vec{B}$ perpendicular to the plane of the semiconductor, in the +y-direction. Assume that the average drift velocity of the moving charges (charge magnitude e) has magnitude v. Figure 1a shows the case of negative charges, such as electrons in N-type semiconductors, and Fig. 1b shows positive charges, Such as holes in P-type semiconductors. The magnetic force is downward whether the moving charges are positive or negative. In both cases the moving charges are driven toward the bottom of the semiconductor by the magnetic force $F_m = evB$.



(a) N-type semiconductor (b) P-type semiconductor

Fig. 1 Forces on charge carriers in a Hall element in a magnetic field

If the charge carriers are electrons, as in Fig. 1a, an excess negative charge accumulates at the lower edge of the semiconductor, leaving an excess positive charge at its upper edge. This accumulation continues until the resulting transverse electrostatic field $\vec{E}_H$ becomes large enough to cause a force $\vec{F}_e$ (magnitude eE_H) that is equal and opposite to the magnetic force $\vec{F}_m$(magnitude evB). After that, there is no longer any net transverse force to deflect the moving charges. This electric field causes a potential difference between opposite edges of the semiconductor, called the Hall voltage U_H.

$$evB = eE_H = e\frac{U_H}{b} \tag{1}$$

The magnitude of the working current is

$$I_s = nevbd \tag{2}$$

where n is the carrier concentration. According to Eq. (2), the average drift velocity of carriers is

$$v = \frac{I_s}{nebd} \tag{3}$$

Eliminating v from Eq. (1) and Eq. (3), we find

$$U_H = \frac{1}{ne}\frac{I_s B}{d} = R_H\frac{I_s B}{d} = K_H I_s B \tag{4}$$

$$B = \frac{U_H}{K_H I_s} \tag{5}$$

That is, the Hall voltage U_H is proportional to I_s and B and inversely proportional to the thickness d of the Hall element. Where the proportional coefficient R_H=1/ne is called the Hall coefficient, which is an important parameter reflecting the strength of the Hall effect of materials. The proportional coefficient K_H=R_H/d=1/ned is called the sensitivity of Hall element, which is equal to the Hall voltage of the Hall element in unit magnetic

field and unit working current. The unit of K_H is V/(A·T). Generally, the larger the K_H, the better.

Eq. (5) is the theoretical basis for measuring magnetic fields by Hall effect. For a certain Hall element, as long as its sensitivity K_H is known and I_s and U_H are measured, the magnitude of magnetic field B can be calculated by Eq. (5).

The carrier concentration n can be obtained from the Hall coefficient or the sensitivity.

$$n = \frac{1}{eR_H} = \frac{1}{edK_H} \tag{6}$$

Due to the high electron concentration n of a metal, its R_H or K_H is small, so it is not suitable for use as a Hall element. In addition, the thinner the thickness d of the element, the higher the K_H. So the method of reducing the thickness is often used to increase sensitivity. However, it cannot be considered that the thinner the thickness is, the better, because the input and output resistances of the element will increase at this time.

According to the force analysis, when the direction of the working current I_s or the magnetic field B changes, the direction of the Hall voltage U_H changes accordingly. If both directions are changed, the polarity of the Hall voltage U_H remains unchanged.

2. The side effect of Hall effect and its elimination

The measured Hall voltage U_H superimposes some additional potential caused by side effects, resulting in system errors. These side effects include

(1) Unequal potential U_0

The two voltage test leads A and A' of the Hall element cannot be welded absolutely symmetrically on both sides of the Hall element (as shown in Figure 2). Even if the magnetic field B=0, because A and A' are not on the same equipotential surface, there is a voltage U_0. $U_0=I_sR_0$, where R_0 is the resistance between the A and A'. The positive or negative of U_0 changes with the direction of I_s.

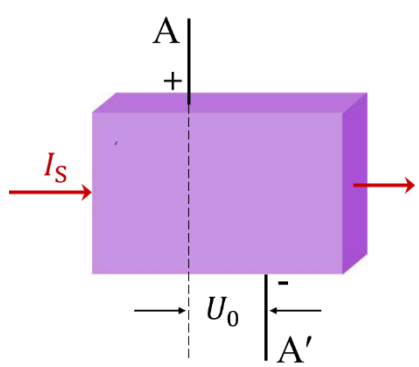


Fig. 2 Cause of unequal potential

(2) Etinghausen effect U_E

The average drift velocity of carriers in Eq. (4) is v, but in reality, the velocities of carriers vary. After the Hall electric field is stabilized, the magnetic force on carriers with a velocity greater than v is greater than the electric field force, the magnetic force on

carriers with a velocity less than v is less than the electric field force. High speed carriers accumulate at one side of the Hall element and the collision with the lattice makes the temperature of this side high. At the other side, there are many low speed carriers accumulated and the temperature is relatively low. As a result, an additional thermoelectric potential U_E between the two sides of the Hall element are generated. Since the direction of the magnetic force on a carrier is related to both the direction of the working current I_s and the magnetic field B, the positive or negative of U_E changes with the direction of I_s or B.

(3) Nernst effect U_N

Because the welding resistances of the working current leads at both ends of the Hall element are not equal, the working current I_s causes a temperature difference between the two ends, resulting in an additional thermoelectric current, which causes an additional potential U_N similar to U_H in the magnetic field. Since the direction of the additional current is determined by the temperature difference between the two ends, the positive or negative of U_N is independent of the direction of the working current I_s, and changes with the direction of the magnetic field B.

(4) Righi-Leduc effect U_R

The carriers of the additional current caused by the Nernst effect have different velocities, so the thermoelectric potential U_R is also caused by Etinghausen effect. Since the direction of the additional current is independent of the direction of the working current, the positive or negative of U_R only changes with the direction of the magnetic field B.

The measured voltage U between A and A' includes not only Hall voltage U_H, but also U_0, U_E, U_N and U_R. Since the positive or negative of the additional voltage is related to the directions of the working current I_s and the magnetic field B (that is, the corresponding excitation current I_M), the impact of some additional potentials can be eliminated by changing the directions of I_s and B. This method of eliminating systematic errors is called symmetrical measurement.

When we set $+I_M, +I_s$ $\quad U_1 = +U_H + U_0 + U_E + U_N + U_R$

$+I_M, -I_s$ $\quad U_2 = -U_H - U_0 - U_E + U_N + U_R$

$-I_M, -I_s$ $\quad U_3 = +U_H - U_0 + U_E - U_N - U_R$

$-I_M, +I_s$ $\quad U_4 = -U_H + U_0 - U_E - U_N - U_R$

We get

$$U_H = \frac{1}{4}(U_1 - U_2 + U_3 - U_4) - U_E \tag{7}$$

In general, $U_H >> U_E$, so U_E can be ignored, then from Eq. (7),

$$U_H \approx \frac{1}{4}(U_1 - U_2 + U_3 - U_4) \tag{8}$$

In general, the Hall voltage U_H is much larger than other additional voltages, so U_1 and U_3 have the same sign, U_2 and U_4 have the same sign, and the two sets of data have the opposite sign, so

$$U_{\mathrm{H}} \approx \frac{1}{4}(U_1 - U_2 + U_3 - U_4) = \frac{1}{4}(|U_1| + |U_2| + |U_3| + |U_4|) \tag{9}$$

Contents

1. The preparation before the experiment.

(1) Turn the knob of working current and excitation current counterclockwise to the end, so that the current is minimum. Set the Hall voltage range of the tester to the high range.

(2) Connect the Hall effect tester with wires.

(3) The Hall element should be preheated for at least 5 min before the experiment. As follows: Disconnect the excitation current switch, working current I_S=5 mA.

2. Measure the Hall element sensitivity K_{H} and calculate the carrier concentration n.

(1) Move the two dimensional moving ruler so that the Hall element is in the center of the electromagnet air gap. Turn on the excitation current switch. Adjust and keep the excitation current I_{M} as 300 mA. Calculate and record the magnetic field B by formula $B = C \cdot I_{\mathrm{M}}$, where C is the coil constant of electromagnet.

(2) Adjust the working current I_{S} from 1.00 mA to 9.00 mA with an increment of 1.00 mA. Measure the corresponding voltage U_{i} (i=1,2,3,4) respectively by changing the reversing switches of the excitation current I_{M} and the working current I_{S}. Calculate and record the corresponding Hall voltage U_{H} in Table 1. Plot the straight line of $U_{\mathrm{H}} - I_{\mathrm{S}}$, obtain the slope K_1.

(3) According to formula $U_{\mathrm{H}} = K_{\mathrm{H}} I_{\mathrm{s}} B$, we can obtain $K_{\mathrm{H}}=K_1/B$. Calculate the carrier concentration n by $n = \frac{1}{edK_{\mathrm{H}}}$, where d is the thickness of the Hall element.

3. Analyse the relationship between the Hall voltage U_{H} and the excitation current I_{M}.

Adjust and keep the working current I_{S} as 3.00 mA. $I_{\mathrm{M}} = 100, 200 \cdots\cdots, 1000$ mA, measure the voltage U_{i} separately and record it in Table 2. Calculate the Hall voltage U_{H}, and use the sensitivity K_{H} of the Hall element obtained from the previous experiment to calculate the magnetic field B according to Eq. (5). Draw the curve of $B - I_{\mathrm{M}}$, and analyze the relationship between the magnetic induction intensity B and the excitation current I_{M}.

4. Measure the intensity and distribution of magnetic field in the air gap of electromagnet.

(1) $I_{\mathrm{M}} = 600\mathrm{mA}$, $I_{\mathrm{S}} = 5.00\mathrm{mA}$. Position the Hall element in the center of the electromagnet air gap in the vertical direction. Adjust the horizontal ruler to the 0 scale and measure the corresponding U_{i}.

(2) Adjust the horizontal ruler to measure the U_{i} at the position given in Table 3.

(3) According to the U_{i} measured above, calculate the Hall voltage U_{H}. Calculate the magnetic field B of each point by Eq. (5). Plot the relationship curve of the magnetic

field B and the horizontal coordinate x, describe the intensity of the magnetic field at different horizontal positions in the air gap of an electromagnet.

Precautions

1. The excitation current I_M is large, so do not connect I_M and I_S wrong, otherwise the excitation current will burn out the Hall element.

2. Hall element and two dimensional moving ruler are easy to break and deform, so they should be avoided from being squeezed or collided.

Data Taking

Table 1 Measurement of $U_H - I_S$ $\quad I_M = 300\text{mA}$

I_s(mA)	U_1(mV)	U_2(mV)	U_3(mV)	U_4(mV)	$U_H = \frac{1}{4}(\lvert U_1\rvert + \lvert U_2\rvert + \lvert U_3\rvert + \lvert U_4\rvert)$ (mV)
	$+I_M$, $+I_s$	$-I_M$, $+I_s$	$-I_M$, $-I_s$	$+I_M$, $-I_s$	
1.00					
2.00					
3.00					
4.00					
5.00					
6.00					
7.00					
8.00					
9.00					

Table 2 Measurement of $B - I_M$ $\quad I_S = 3.00\text{mA}$

I_M(mA)	U_1(mV)	U_2(mV)	U_3(mV)	U_4(mV)	U_H(mV)	B (mT)
	$+I_M$, $+I_s$	$-I_M$, $+I_s$	$-I_M$, $-I_s$	$+I_M$, $-I_s$		
100						
200						
300						
400						
500						
600						
700						
800						
900						
1000						

Table 3 Measurement of $B - x$ $\quad I_M = 600\text{mA}, I_S = 5.00\text{mA}$

x(mm)	U_1(mV)	U_2(mV)	U_3(mV)	U_4(mV)	U_H(mV)	B (mT)
	$+I_M$, $+I_s$	$-I_M$, $+I_s$	$-I_M$, $-I_s$	$+I_M$, $-I_s$		
0						
2						
4						
6						
8						
10						
12						
15						
20						

25						
30						
35						
40						
45						
48						
50						

Data Analysis

1. According to the data in Table 1, plot the straight line of $U_{\mathrm{H}} - I_{\mathrm{S}}$, and obtain the slope K_1.

2. Calculate the sensitivity of Hall element $K_{\mathrm{H}}=K_1/B$. (Magnetic field B is obtained by the formula $B=C\cdot I_{\mathrm{M}}$)

3. Calculate the carrier concentration n by $n = \frac{1}{edK_{\mathrm{H}}}$.

4. Using the sensitivity K_{H} of the Hall element, calculate the magnetic field B in Table 2 and Table 3 according to Eq. (5).

5. According to the data in Table 2, draw the curve of $B - I_{\mathrm{M}}$, and analyze the relationship between the magnetic field B and the excitation current I_{M}.

6. According to the data in Table 3, plot the relationship curve of the magnetic field B and the horizontal coordinate x, describe the intensity of the magnetic field at different horizontal positions in the air gap of an electromagnet.

Questions

1. A large self-induced electromotive force is generated when the excitation current changes direction, which may burn out the power supply. How to operate to avoid generating large self-induced electromotive force?

2. When measuring magnetic field with Hall element, if the direction of the magnetic field is not parallel to the normal direction of the element, will the measured value of the magnetic field be larger or smaller?